Completion of the Hadamard Expansion of I_B in Two Dimensions

Robert W. McGwier
CTO, Cohere Technology Group
GitHub: n4hy

15 August 2026 — Version 1

Abstract

The multi-digit coefficient of the kinematic residue I_B was left open pending an analytic Hadamard expansion of the current two-point function under modular flow. That expansion is completed here in $d = 2$.

The identity

$$\frac{1}{\sinh^2 w} = \sum_{n\in\mathbb{Z}} \frac{1}{(w - in\pi)^2}$$

is exact off the poles $w \in i\pi\mathbb{Z}$. The $n = 0$ term is the coincidence singularity. The Hadamard-finite remainder is the $n \neq 0$ sum, equivalently $1/\sinh^2 w - 1/w^2$. Folding against the modular kernel reduces to a polygamma kernel $H(\tau)$ derived below. The formula for the building block $I_{\rm sing}(\zeta)$ matches an independent sech^2 quadrature to 2×10^{-13}.

$H(\tau)$ is *not* proportional to $1/\tau^2$, to $1/\sinh^2\tau$, to $I_{\rm sing}(\tau)$, or to $1/\sinh^2\tau - 1/\tau^2$. Relative residues formed with four Gaussian sources therefore move with the source. There is no single universal multi-digit coefficient of I_B independent of β.

The Final Status label of I_B stays **[O]** if "coefficient" means one number. The expansion itself is closed.

Rule

This note distinguishes an exact identity, a machine-checked special function formula, a numerical spread, and an admission. It does not upgrade a Final Status **[O]** to a derived number that the algebra does not contain.

1 Setup

On the interval $(-R, R)$ with $R = 1$, modular flow is $x(s) = \tanh(\operatorname{artanh} x + \pi s)$. After $x = \tanh u$ the weight-one Jacobians cancel [1, 3] and

$$Q(s) = \int d\tau\, C(\tau)\, \frac{1}{\sinh^2(\tau + \pi s)}, \qquad C(\tau) = \int du\, b(u)b(u - \tau), \quad b(u) = \beta(\tanh u). \tag{1}$$

The kernel is $K(s) = \pi/(2\cosh^2(\pi s))$, $\int K = 1$, and

$$\int K(s)\, f(\tau + \pi s)\, ds = \int \frac{d\alpha}{2\cosh^2\alpha}\, f(\tau + \alpha). \tag{2}$$

2 The complete expansion

Theorem 2.1 (Mittag–Leffler)**.** *For $w \notin i\pi\mathbb{Z}$,*

$$\frac{1}{\sinh^2 w} = \sum_{n=-\infty}^{\infty} \frac{1}{(w - in\pi)^2}. \tag{3}$$

This is the Hadamard series: every pole is written, none is omitted. The coincidence is $n = 0$. The finite remainder

$$h(w) := \frac{1}{\sinh^2 w} - \frac{1}{w^2} = \sum_{n \neq 0} \frac{1}{(w - in\pi)^2} \tag{4}$$

is entire in a neighbourhood of the real axis and equals $-1/3$ at $w = 0$.

The Hadamard-finite smeared correlator is therefore

$$Q_H(s) = \int d\tau\, C(\tau)\, h(\tau + \pi s) = \sum_{n \neq 0} \int d\tau\, \frac{C(\tau)}{(\tau + \pi s - in\pi)^2}. \tag{5}$$

3 The modular integral of the finite part

Define

$$I_{\text{sing}}(\zeta) := \frac{1}{2} \int_{-\infty}^{\infty} d\alpha\, \frac{\operatorname{sech}^2 \alpha}{(\alpha + \zeta)^2}. \tag{6}$$

Residues of sech^2 at $i\pi(n + \frac{1}{2})$ give, for $\operatorname{Im} \zeta > 0$,

$$I_{\text{sing}}(\zeta) = \pi^{-2}\, \psi^{(2)}\left(\tfrac{1}{2} - \frac{i\zeta}{\pi}\right), \tag{7}$$

and $I_{\text{sing}}(-\zeta) = I_{\text{sing}}(\zeta)$. On the real axis $I_{\text{sing}}(\tau) = \pi^{-2} \operatorname{Re} \psi^{(2)}(\frac{1}{2} - i\tau/\pi)$.

Machine check (`m9_4_ib_hadamard_complete.py`): (7) versus sech^2 quadrature at three complex points, maximum relative error 1.99×10^{-13}.

The finite modular integral is the kernel

$$\begin{aligned} H(\tau) &:= \int K(s)\, h(\tau + \pi s)\, ds = \sum_{n \neq 0} I_{\text{sing}}(\tau - in\pi) \\ &= \pi^{-2} \sum_{m=1}^{\infty} \Big[\psi^{(2)}\big(m + \tfrac{1}{2} + \tfrac{i\tau}{\pi}\big) + \psi^{(2)}\big(m + \tfrac{1}{2} - \tfrac{i\tau}{\pi}\big)\Big]. \end{aligned} \tag{8}$$

Then

$$\int K(s)\, Q_H(s)\, ds = \int d\tau\, C(\tau)\, H(\tau). \tag{9}$$

4 There is no universal ratio

A single coefficient would mean $H = \lambda K_{\text{loc}}$ for some local kernel K_{loc} independent of C. On $\tau \in \{0.3, 0.5, 0.8, 1.0, 1.5, 2.0, 3.0\}$ the ratio H/K_{loc} has spread

K_{loc}	spread of H/K_{loc}
$I_{\mathrm{sing}}(\tau)$	0.553
$1/\tau^2$	0.968
$h(\tau)$	0.210
$1/\sinh^2\tau$	not constant

Relative residues $r = \int CH / \int CK_{\mathrm{loc}}$ for Gaussians of width $\sigma \in \{0.4, 0.7, 1.0, 1.4\}$:

denominator	mean r	spread in σ
I_{sing}	0.0585	7.08×10^{-2}
h	0.898	4.49×10^{-2}
$1/\tau^2$	-0.0203	1.64×10^{-2}
$1/\sinh^2$	-0.0208	1.71×10^{-2}

None of these spreads is a rounding error.

Admission. The Hadamard series is complete. What it does *not* produce is one multi-digit number independent of β. Asking for that number was slightly the wrong question: the finite object is the kernel $H(\tau)$, and H is not a multiple of a local quadratic form. The Final Status line "multi-digit coefficient of I_B" remains **[O]** as a *number*. As an expansion, the $d = 2$ problem is closed.

5 What remains

- $d = 4$ balls, where the current two-point function is not $1/(x-y)^2$ on a line and CHM is a conformal image of Rindler, not a Möbius map on $\mathbb{R}$.
- A renormalization convention that would force a unique λ by extra physical input. None is supplied by $K(s)$ alone.
- Digits of $H(\tau)$ at a conventional τ are available from (8) to any precision; they are values of a function, not a coefficient of I_B.

Script

`m9_4_ib_hadamard_complete.py`, data `m9_4_ib_hadamard_complete.json`.

References

[1] R.W. McGwier, "Evaluation of the Universal Kinematic Integral for Condition NL," https://github.com/n4hy/New_Model_Emergent_Gravity.

[2] R.W. McGwier, "Status of the High-Precision Coefficient of I_B (v4)," ibid.

[3] R.W. McGwier, "Analytic Attempt at the Coefficient of I_B," ibid.

[4] H. Casini, M. Huerta and R.C. Myers, JHEP 05 (2011) 036.

[5] H. Casini, E. Teste and G. Torroba, J. Phys. A 50 (2017) 364001.